Coursework Assessment 2 Fquency Domain(Must complete **2 topics** or more. One or more from 1-3 and one or more from 4-7)

Signal and System Time Domain Analysis MATLAB Simulation

Section 1

Topic 1: Analysis of Instrument Timbre Characteristics

Timbre is the most crucial and essential attribute of an instrument's sound. When pitch and loudness are the same, people often rely on timbre to distinguish the sounds of different instruments first. Every instrument has its unique timbre. From a physical perspective, the timbre characteristics of different instruments vary because the vibration methods of different instruments are not the same. These vibrations consist of multiple overtones. Different combinations of overtones determine the timbre of a particular instrument.

On a technical level, timbre depends on the waveform of the sound wave; from the perspective of signal analysis, the waveform in turn depends on its spectral structure. According to the Fourier transform, any periodic signal can be decomposed into a sum of sinusoidal components with different frequencies and amplitudes. Among these, the component with the lowest frequency is called the fundamental frequency, which determines the pitch; components whose frequencies are integer multiples of the fundamental frequency are called harmonics (referred to as overtones in music), and their relative amplitudes determine the shape of the waveform, thereby determining the timbre. Therefore, timbre can be technically described as the amplitude distribution relationship between the fundamental frequency and its harmonics. The differences in timbre among different musical instruments are essentially differences in this distribution relationship.

The timbral characteristics of musical instruments have important analytical value. Using spectral features to characterize the timbre of instruments essentially builds a bridge between digital signal processing technology and auditory perception. This method provides an important basis for computer analysis or the synthesis of instrument sounds. On this foundation, it is possible to further achieve automatic recognition of instrument types, intelligent music retrieval, and recommendations. In audio engineering, this method also offers a scientific basis for mixers, allowing targeted equalization and sound field reconstruction based on the harmonic distribution of

instruments.

In this experiment, based on the collected sound signals of the violin, trumpet, clarinet, and flute (stored respectively in the binary files 'xtq.dat', 'xh.dat', 'dhg.dat', and 'cd.dat'), a program is written to analyze the sound signals of the four instruments and plot the corresponding waveforms and spectra. By observing the fundamental frequency, harmonic distribution characteristics, and harmonic amplitude decay trends in the spectra, the timbre characteristics of different instruments can be explained.

(1) Collect acoustic signals of violin, trumpet, clarinet and flute, and save them as binary files: xtq.dat, xh.dat, dhg.dat, cd.dat respectively.（*If it is possible to collect signals, this part can be completed; otherwise, the provided data can be used.*）

(2) Plot the time-domain waveforms of the four instrumental acoustic signals and perform spectrum analysis.

(3) Summarize the spectral characteristics of different instrumental sounds.

Topic 2: Instrument Recognition

In previous experiments, frequency spectrum analysis was conducted on the sound signals of four instruments: violin, trumpet, clarinet, and flute, to observe indicators such as the fundamental frequency positions, harmonic decay trends, proportions of odd harmonics, and effective number of harmonics of different instrument sound signals, thereby summarizing their characteristic patterns.

Specifically, it can be summarized as follows:

The violin's string vibration generates a complete series of integer multiple harmonics, and the low-order harmonic energy is abundant; this characteristic originates from the physical laws of string vibration and is stable.

The clarinet's reed vibration causes the odd harmonics to be significantly stronger than the even harmonics; this feature arises from the asymmetry of the boundary conditions and is a typical marker of reed instruments.

The trumpet's high-order harmonics are prominent, originating from the high-frequency resonance characteristics of the metal tube body.

The flute has fewer harmonics and rapid decay, originating from specific excitation patterns of the air stream at the embouchure edge.

By summarizing the spectral patterns of four types of musical instruments, a simple 'instrument feature database' can be established; in this experiment, it is required to use the feature database to extract the spectral indicators of an unlabelled signal, compare them with the existing patterns, and ultimately provide a discrimination

conclusion. This process is essentially a prototype of pattern recognition and represents a further application of musical instrument sound signal analysis in real-life situations.

This experiment requires plotting the spectrum of a given sound signal; observing features such as the fundamental frequency, number of harmonics, harmonic distribution, and harmonic attenuation amplitude, and then, based on the summarized patterns, determining which instrument performed the given signal.

(1) Plot the waveform of an unknown acoustic signal and conduct spectrum analysis.

(2) Observe its fundamental frequency, harmonic distribution, harmonic amplitude attenuation trend and other features.

(3) Identify which instrument produces the signal.

Topic 3:Musical Instrument Mixture Separation

In the two experiments, "Analysis of Instrument Timbre Characteristics" and "Instrument Classification," we can summarize the following conclusions: Different instruments have significant differences in harmonic structures; the violin has abundant low-order harmonic energy, the clarinet is dominated by odd harmonics, the trumpet highlights high-order harmonics, and the flute has fewer harmonics that decay quickly. These differences result in distinct "fingerprints" in the frequency domain. Theoretically, when sound propagates through air, the sound pressure waves generally satisfy the principle of linear superposition. Therefore, when multiple instruments play simultaneously, the time-domain signal measured at a point can be approximately regarded as the algebraic sum of the signals of each instrument. According to the properties of the Fourier transform, for two real signals $x_1(t)$ and $x_2(t)$, we have：

$$F\left\{x_1(t)+x_2(t)\right\}=X_1(f)+X_2(f) \tag{2-1}$$

That is, the spectrum of the summed signals equals the sum of the spectra of the individual signals.

From the above principle, it can be inferred that the spectrum of a mixed signal of different musical instruments is roughly equal to the linear superposition of the spectra of each instrument. This provides a basis for analyzing the mixed signals of musical instruments through spectral analysis.

This experiment requires plotting the spectrum of a sound signal composed of the mixture of two types of musical instruments, observing the fundamental frequency and harmonic distribution in the spectrum. Based on the characteristics of musical instruments obtained from previous experiments, analyze and explain the results, and try to identify which instruments are included in the mixed signal.

(1) Use MATLAB to plot the waveform of a mixed acoustic signal and perform spectrum analysis.

(2) Observe the fundamental frequency and harmonic distribution in the spectrum, and identify the instruments contained in the mixture.

(3) Compare the spectrum of the mixed signal with those of single instruments, analyze the patterns and explain the mechanism.

Section 2

Topic 4: Speech Signal Denoising

Frequency analysis of speech signals reveals fundamental frequencies and harmonics. However, environmental noise is inevitable in real-world acquisition.

When noise is strong, the time-domain waveform of speech is obscured, blurring its periodic structure. Proper denoising enhances effective speech components and is critical for voiceprint recognition.

Wavelet threshold denoising is the most widely used method, including wavelet transform, thresholding of wavelet coefficients, and signal reconstruction via inverse wavelet transform.

Specify a basic function $\varphi(t)$, perform translation on *b*, then carry out scaling at different scales, and compute the inner product with the signal *x*(*t*) to be processed, obtaining the following formula:

$$WT_X\left(a,b\right)=\frac{1}{\sqrt{a}}\int x(t)\varphi(\frac{t-b}{a})dt=\int x\left(t\right)\varphi_{ab}(t) \tag{2-2}$$

Where *a* is the dilation factor and *b* is the translation factor.， $\varphi_{ab}(t)$ is the wavelet basis function. In the formula, the independent variables a and b are continuous variables, so this formula is called the Continuous Wavelet Transform (CWT). The dilation factor a is used to perform scaling transformation on the wavelet basis function. By changing the value of the dilation factor a ,the wavelet basis function $\varphi_{ab}(t)$ can be extended or shortened to obtain wavelet coefficients of different frequency layers after transformation. The translation factor *b* is used to adjust the position of the wavelet basis function. By changing the value of the translation factor *b*, the wavelet basis function $\varphi_{ab}(t)$ can be shifted in the time domain of the signal *x*(*t*) to be processed, so as to perform computation with the original signal at different time positions.

When analyzing signals using a computer, the signals *x*(*t*) must be discrete. Therefore, during computer execution, the variables in the continuous wavelet transform formula must take discrete values, and the discrete wavelet transform has emerged accordingly. The discretization method for the variables a is as follows:

$$a=a_o^j\ (j\in Z,a_o>1) \tag{2-3}$$

where a_0 is usually taken as 2.

The discretization method for *b* is as follows:

$$b = kb_o a_o^j \left(j \in Z, b_o \geq 1 \right) \tag{2-4}$$

where b_0 is usually taken as 2.

Substituting the discrete forms of the dilation factor a and translation factor b into the continuous wavelet transform formula yields the following expression:

$$\varphi_{j,k}(t) = \frac{1}{\sqrt{a_o^j}} \varphi\left(\frac{t - kb_o a_o^j}{a_o^j} \right) = a_o^{\frac{-2}{j}} \varphi\left(a_o^{-j} t - kb_o \right) \ldots\ldots j, k \in Z \tag{2-5}$$

Upon further simplification, the expression for the Discrete Wavelet Transform (DWT) can be obtained:

$$WT_x(j,k) = \int x(t) \varphi_{ab}^*(t) dt \tag{2-6}$$

In practical applications, we usually pay more attention to the low-frequency approximation coefficients and high-frequency detail coefficients obtained from the discrete wavelet transform. Therefore, based on the theory of discrete wavelet transform, Stephane Mallat proposed the Mallat pyramid algorithm in 1989. Using this method, the most noteworthy approximation coefficients and detail coefficients in the wavelet transform can be obtained directly. The calculation method for the approximation coefficients is as follows:

$$a_{j+1}(n) = \sum_k x(k) \bullet h(n - 2k) \tag{2-8}$$

where $h(n)$ is the low-pass filter constructed using the wavelet basis function.

The calculation method for the detail coefficients is as follows:

$$d_{j+1}(n) = \sum_k x(k) \bullet g(n - 2k) \tag{2-9}$$

where $g(n)$ is the high-pass filter constructed using the wavelet basis function.

The proposal of the Mallat pyramid algorithm accelerates the process of signal processing using wavelets, greatly facilitating the implementation of discrete wavelet transform on computers.

After performing wavelet transform on the original signal, high-frequency coefficients and low-frequency coefficients can be obtained. Thresholding the wavelet coefficients is the main method to realize wavelet threshold denoising. Generally, noise exists in the high-frequency coefficients, so wavelet threshold denoising only processes the high-frequency coefficients with thresholding. Common thresholding methods mainly include hard thresholding and soft thresholding. The VisuShrink universal threshold used for denoising is calculated as follows.

$$\lambda_i = \sigma_i \sqrt{2 \log_2(n_i)} \tag{2-10}$$

where n_i is the length of the wavelet coefficients in the i-th level high-frequency coefficients of the input signal, and σ_i is the overall variance of the i-th level coefficients. A commonly used shortcut calculation form σ_i is as follows:

$$\hat{\sigma}_i = MAD_i / 0.6754 \tag{2-11}$$

MAD_i is the absolute median of the i-th level high-frequency wavelet coefficients.

The hard threshold method is simple and straightforward: wavelet coefficients smaller than the threshold are set to zero, while those larger than the threshold remain unchanged.

The formula for the hard threshold method is as follows:

$$\hat{x} = T_i(Y,t) = \begin{cases} Y & |Y| \geq \lambda \\ 0 & |Y| < \lambda \end{cases} \tag{2-12}$$

Where Y is the wavelet coefficient of the i-th level, λ is the threshold for the i-th level wavelet coefficients.

The soft threshold method further processes wavelet coefficients larger than the threshold by setting them to the difference between the coefficient and the threshold. Its formula is as follows:：

$$\hat{x} = T_h(Y,t) = \begin{cases} \mathrm{sgn}(Y)(|Y| - t) & |Y| \geq t \\ 0 & |Y| < t \end{cases} \tag{2-13}$$

Compared with the hard threshold method, the wavelet coefficients processed by the soft threshold method have better overall continuity, and most of the detailed information can be preserved.

(1) Collect male and female speech signals and save them as .dat data files. （*If it is possible to collect signals, this part can be completed; otherwise, the provided data can be used.*）

(2) Use MATLAB to plot the waveforms of the collected speech signals and perform spectrum analysis.

(3) Perform wavelet threshold denoising on the male and female speech signals.

(4) Compare the results before and after denoising to understand the significance of denoising in signal analysis.

Topic 5: Biomedical Signal Characteristics

The generation mechanism of heart sound signals is as follows: vibrations formed by the heart valves and large blood vessels under the impact of blood flow; vibrations produced by turbulence and eddy currents resulting from the acceleration and deceleration of blood flow within the heart, as well as their effects on the heart valves, atria and ventricular walls; and vibrations formed by the rapid increase and decrease in the rigidity of the myocardium under periodic electrical cardiac activities. These vibrations propagate to the body surface through the cardiothoracic transmission system, forming the body surface heart sound.

Heart sound is one of the relatively complex physiological signals. Heart sounds originate from vibrations of multiple vibration sources at different spatial positions and different times. Non-negligible interactions exist between these vibration sources, and each source exhibits fairly strong time-varying characteristics.

The periodic beating of the heart renders heart sounds periodic, and the waveform is shown in Figure 9-1. For the diagnosis of heart diseases, the key lies in correctly dividing and identifying the boundaries of each component of the heart sound, namely heart sound segmentation. A common method is to identify the first heart sound S1 and the second heart sound S2 according to the regularity of systole and diastole, and then screen out abnormal heart sounds based on the segmentation results to complete medical diagnosis.

Figure 2.1 Phonocardiogram Signal Waveform

The generation mechanism of pulse signals mainly includes the pumping action of the heart and the elastic response of arterial blood vessels. The heart is the power source for pulse generation. When the heart ventricles contract, blood is pumped into the aorta and pulmonary artery. Taking the left ventricle as an example, during systole, the left ventricular myocardium contracts violently, causing a sharp rise in intraventricular pressure. When the intraventricular pressure exceeds the aortic pressure, the aortic valve opens and blood is rapidly pumped into the aorta. The sudden impact and filling

of blood rapidly increase the pressure and volume within the aorta, generating the initial driving force of the pulse. This is similar to a water pump forcibly pumping water into a pipe, where the impact of the water flow on the pipe initiates the pulse signal.

The aorta and arteries at all levels are elastic. When the heart pumps blood and raises the arterial pressure, the arterial walls expand accordingly to accommodate the increased blood volume. This elastic expansion is analogous to the inflation of a balloon. During diastole, the ventricles stop pumping blood, and the arteries rely on their own elastic recoil to propel blood forward continuously, maintaining continuous blood flow in the vessels. This also causes periodic changes in arterial pressure and volume, forming the pulse wave.

The pulse signal is also a periodic signal. A single-cycle pulse waveform is shown in Figure 2.2, which mainly consists of a rising branch and a falling branch. Based on the processes of cardiac pumping and blood propagation in blood vessels, the pulse wave can be divided into seven characteristic points: *a, b, c, d, e, f,* and g. Among them, segment *bc* is the rising branch, during which the aorta opens and the heart pumps blood into the aorta, increasing aortic pressure and raising the pulse curve to point *c*, the peak of the entire pulse wave. Subsequently, blood is pumped out of the left atrium, forming the tidal wave segment *de*, with point *d* as the trough of the tidal wave. Afterwards, the right atrium opens and blood rebounds to form a trough at point *f*. Point *g* is the peak of the dicrotic wave, caused by cardiac diastole. The entire pulse period is T, representing one complete cardiac contraction and diastole. T_c, T_d, and T_g represent the time to reach the corresponding characteristic points, respectively, and h_c, h_e, and h_g represent the amplitudes of the main peak, tidal wave, and dicrotic wave, respectively. Characteristic points of the pulse are widely used in medical diagnosis; for example, segment b_f represents cardiac systole, and segment $f_{b'}$ represents cardiac diastole.

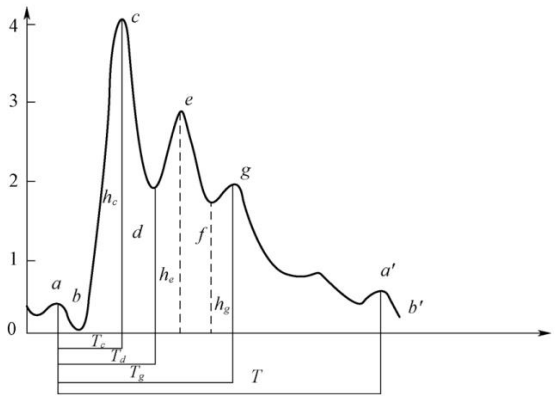

Figure 2.2 Pulse Signal Waveform

The mechanism of lung sound generation mainly includes three aspects: turbulence produced by gas flow, vibration of airway walls, and activities of alveoli and pleura.

During normal respiration, gas flows through the respiratory tract. When the gas flow velocity is sufficiently high, local stenosis occurs in the airway, or branching appears, turbulence will be generated. For example, at the bifurcation of the main bronchi, the direction of airflow changes, which easily induces turbulence. Such turbulence causes vibrations of the airway walls and surrounding tissues, thereby producing sounds. During deep breathing, the gas flow rate increases, and the airflow velocity accelerates, which raises the possibility and intensity of turbulence, further altering characteristics of lung sounds such as volume and frequency.

Airway walls including the trachea and bronchi exhibit certain elasticity. When airflow impacts the airway walls, or surrounding tissues such as smooth muscles contract and relax, the airway walls vibrate. For instance, during coughing, the intercostal muscles and diaphragm contract violently, causing a sharp rise in intrathoracic pressure. Airflow rushes out of the airway rapidly and impacts the airway walls strongly, producing a relatively loud cough sound. Physical properties of the airway walls, such as thickness and elasticity, also affect lung sound generation. For example, when airway walls become stiff or thickened due to inflammation, their vibration characteristics change accordingly, resulting in different lung sounds.

Alveoli expand and contract during respiration. During inspiration, alveoli expand and gas enters; during expiration, alveoli contract and gas is expelled. Elastic fibers in the alveolar walls produce slight vibrations during this process. Vibrations also arise when pressure changes within alveoli are uneven or tension differs between adjacent alveoli. The pleura is a thin membrane covering the surface of the lungs and lining the thoracic cavity. Under normal conditions, a small amount of fluid in the pleural cavity acts as a lubricant. During respiratory movement, expansion and contraction of the lungs cause friction and vibration of the pleura. For example, in diseases such as pleurisy, the pleura becomes rough. Friction between the pleurae intensifies during breathing, producing friction sounds and leading to changes in lung sounds.

Lung sound signals originate from complex interactions between elastic biological

tissues and airflow or fluids. These subtle vibrational changes contain abundant medical information. Lung sound signals are weak, complex, variable random signals, and the waveform is shown in Figure 2.3.

Figure 2.3 Lung Sound Signal Waveform

(1) Collect three kinds of human physiological signals: heart sound, pulse and lung sound, and save them in DAT format. （*If it is possible to collect signals, this part can be completed; otherwise, the provided data can be used.*）

(2) Draw the waveforms of the three signals and perform spectrum analysis.

(3) Summarize the characteristics of the three human physiological signals.

(4) Observe the spectra of heart sound and pulse signals, and summarize the relationship between them.Further compare the waveform and spectrum characteristics of the three physiological signals, and try to explain the differences between lung sound signals and heart sound and pulse signals.

Topic 6: Comparison between Heart Sound Signal and ECG Signal

An electrocardiogram (ECG) signal is a bioelectrical signal that reflects the electrical activity of the heart. Its generation mechanism is as follows: the sinoatrial node of the heart acts as the pacemaker and spontaneously generates electrical impulses. The impulses propagate along the atria to the atrioventricular node, and then to the ventricles through the atrioventricular bundle and the left and right bundle branches, triggering the depolarization and repolarization of myocardial cells. The potential changes of myocardial cells during the entire process are acquired by body surface electrodes, forming the ECG signal.

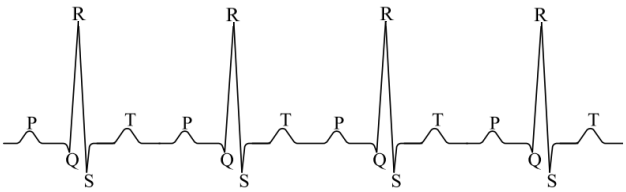

Figure 2.4 ECG Signal Waveform

The ECG signal is also a periodic signal, and its waveform is shown in Figure 2.4. The waveform of an ECG signal consists of a *P* wave, a QRS complex, and a *T* wave. Among them, the *P* wave corresponds to the process of atrial excitation, with a gentle waveform and small amplitude; the QRS complex corresponds to ventricular excitation, with a sharp waveform and the largest amplitude, making it the most recognizable part of the ECG signal; the *T* wave corresponds to ventricular repolarization, with a waveform similar to the *P* wave and an amplitude between that of the *P* wave and the QRS complex.

(1) Plot the time-domain waveform of the ECG signal and perform spectrum analysis on it.

(2) Combined with the results of Experiment 5, compare the waveforms and frequency spectra of the heart sound signal and the ECG signal, and explain the relationship and differences between them.

Topic 7: Spectrum Analysis of Pulse Signal Spectrum

For a given time-domain signal $x(t)$, its power spectrum can be obtained by performing Fourier transform:

$$S_x(f) = \left| F\{x(t)\} \right|^2 \tag{2-14}$$

$F\{\cdot\}$ denotes the Fourier transform operator, Taking the Fourier transform of the logarithm of the power $S_x(f)$ spectrum yields:

$$c(q) = \left| F\{\log S_x(f)\} \right| \tag{2-15}$$

where log represents logarithmic weighting. This equation represents a further spectrum analysis of the frequency spectrum, which is the concept of the cepstrum. The independent variable q is referred to as the quefrency. Small values q correspond to low quefrencies, representing rapid fluctuations and dense harmonics on the spectrum; while large values q correspond to high quefrencies, representing slow fluctuations and sparse harmonics on the spectrum.

The cepstrum is the spectrum of the spectrum. By taking the logarithm of the signal power spectrum and then performing the Fourier transform, periodic components in the complex spectrum can be analyzed. When spectral lines with a period of f_0 exist in the power spectrum of the signal, according to the property of the Fourier transform that periodicity in the frequency domain corresponds to impulses in the time domain, this periodic component will exhibit peaks at $q = 1/f_0$ and its integer multiples（$q = 2/f_0, 3/f_0, \ldots$）in the cepstrum domain. By extracting the quefrencies corresponding to these peaks, the periodic frequency components in the power spectrum can be obtained.

The purpose of logarithmic weighting is to amplify peaks of different magnitudes in the spectrum by different factors, so as to improve the accuracy of transformation. Meanwhile, it has a deconvolution effect, facilitating the separation and extraction of target signals. If a signal is formed by the convolution of two signals, its cepstrum is the sum of the cepstra of the two signals. If the two components in the convolved signal are separable in the quefrency domain, the spectral amplitude of the original signal can be recovered through inverse transformation via quefrency domain weighting (i.e., cepstral filtering).

The generation mechanism of the pulse signal mainly includes the pumping action of the heart and the elastic response of arterial blood vessels. The heart is the power source of pulse generation. When the ventricles of the heart contract, blood is pumped into the aorta and pulmonary artery, thereby generating the initial driving force of the pulse. The pulse signal is a periodic signal whose frequency spectrum presents a discrete spectrum, and the corresponding cepstrum appears as a series of impulse-like

spectral lines. The pulse wave can be regarded as the convolution result of the source signal of cardiac beating and the physiological transmission path system of the human body. It is difficult to separate them in the time domain and frequency domain. However, in the quefrency domain, the cepstrum peaks representing the cardiac cycle and those reflecting the characteristics of the transmission path are usually far apart on the quefrency axis and easy to separate. Therefore, the two can be effectively separated in the quefrency domain by cepstral filtering, so as to extract the cardiac source signal and the impulse response of the physiological transmission path respectively.

(1) Draw the waveform of the pulse signal and perform spectrum analysis.
(2) Based on the results of spectrum analysis, explain the periodicity of the pulse signal.
(3) Perform cepstrum analysis on the pulse signal and interpret the results.
(4) Perform inverse Fourier transform on the cepstrum of the pulse signal, and explain the physical meaning of the results.